

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations

May - June 2023

20MEE11 – FINITE ELEMENT METHOD

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

1. a) Write any four advantages of FEM.
- b) Explain plane stress and plane strain conditions with examples.
- c) For the spring shown in Fig. 1c, determine the nodal displacements using principle of minimum potential energy.

Marks	BT*	CO*	PO*
4	L*1	1	1
6	L2	1	1

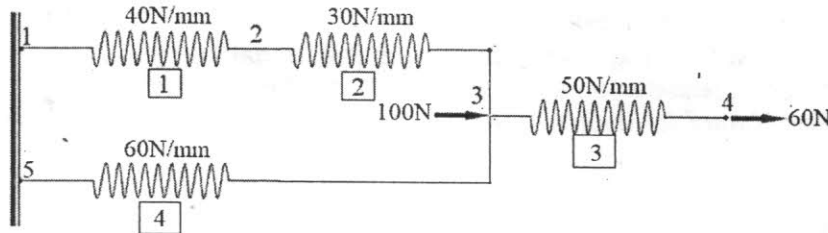


Fig. 1c

10	L3	2	2
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2. a) Derive the stiffness matrix for 1 dimensional bar element using direct stiffness method.
- b) Determine the displacement at the midpoint of the bar shown in fig 2(b) using Rayleigh Ritz method. Take $E=70\text{GPa}$, $A=100\text{mm}^2$.

10	L3	1	2
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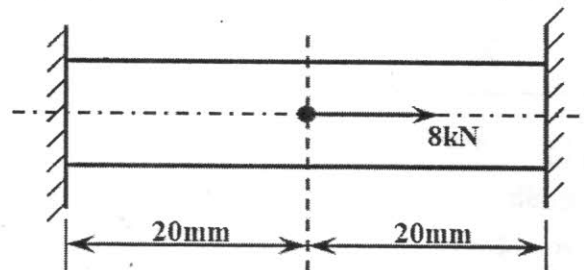


Fig. 2b

10	L3	2	2
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3. a) Sketch the different types of 2D and 3D elements used in FEM.
- b) Derive the 2D relationship between stress and strain components.
- c) Using Galerkin's method find the displacement of a bar subjected to point load P at the free end having area A , length L and Young's Modulus E .

4	L2	1	1
6	L3	1	2

10	L3	2	2
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Unit – II

4. a) Explain with example how the Pascal's triangle is used for choosing terms for displacement polynomial for higher order triangular elements.
- b) Define convergence. Explain h and p method of convergence.
- c) Derive the expression for elemental stiffness matrix for a truss element.
5. a) Derive the shape functions for a 1D linear 2 noded horizontal element in natural coordinate system.

5	L3	3	1
7	L2	3	1

8	L3	4	2
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10	L3	3	2
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P.T.O.

- b) Determine the nodal displacements and stresses in each element for the bar shown in Fig. 5(b) due to applied force $P=100\text{kN}$ using elimination approach.

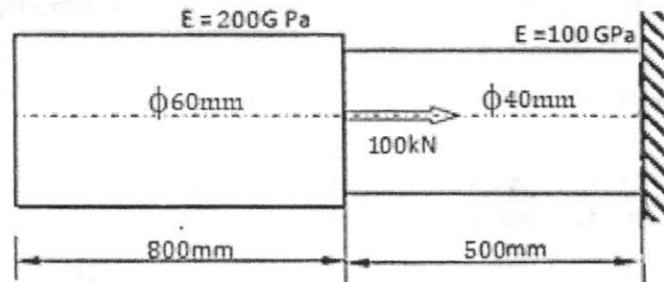


Fig. 5b

10 L3 4 2

6. a) What do you mean by displacement model? State the reasons why polynomial displacement models are widely used? What are the considerations for selection of order of polynomial?
- b) For the two bar truss shown in Fig. 6(b), determine the nodal displacements and element stresses subjected to force $P=1000\text{kN}$. Take $E=210\text{GPa}$ and $A=600\text{mm}^2$ for each element.

8 L1 3 1

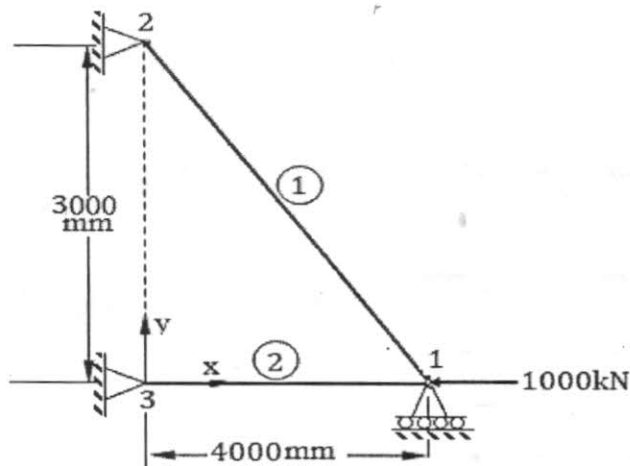


Fig. 6b

12 L3 4 2

Unit - III

7. For the beam shown in Fig. 7, determine the slopes and deflection at the mid span taking $E=200\text{GPa}$ and $I=4 \times 10^6\text{mm}^4$.

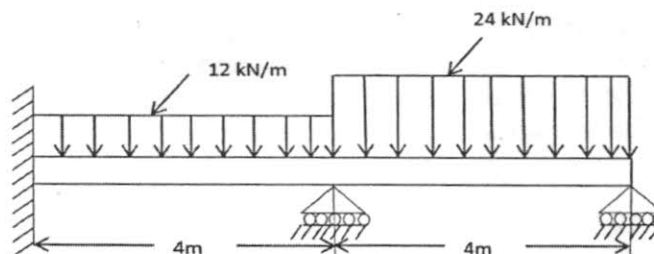


Fig. 7

20 L3 5 2

8. Evaluate the element stiffness matrix for a CST element considering plane strain condition whose details are as follows:
 $(x_1, y_1) = (50, 60)$, $(x_2, y_2) = (80, 90)$ & $(x_3, y_3) = (30, 75)$
 Element thickness = 6 mm
 Poisson's Ratio = 0.3
 Young's Modulus = 200 GPa

20 L3 5 2